

JANSATHI: PROJECT BUILD REPORT

A Comprehensive Case Study on Civic Intelligence & Municipal Operations

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1. PROJECT OVERVIEW

Jansathi is a civic engagement and municipal operations platform designed to bridge the communication gap between everyday citizens and urban municipal departments. It simplifies how local public infrastructure problems—such as hazardous potholes, broken streetlights, overflowing waste dumps, and water pipeline leaks—are reported, prioritized, and resolved.

Jansathi converts unstructured citizen complaints into geotagged, AI-triaged action items for municipal workers, while providing citizens with transparent tracking and community recognition.

Target Users:

- **Citizens and Neighborhood Volunteers:** Residents seeking a reliable, transparent way to report civic problems without bureaucratic delays or lost complaints.
- **Municipal Officers and Field Dispatchers:** City engineers and command center personnel who need structured incident queues, duplicate report consolidation, and urgency-based crew dispatching.

2. THE PROBLEM

Urban civic administration in growing cities frequently suffers from three clear issues:

- **Lack of Feedback on Complaints:** Traditional complaint portals often feel like a black hole. Once a report is filed, citizens rarely receive updates, estimated resolution timelines, or proof of work, leading to frustration and apathy.
- **Triage and Routing Overload:** Municipal officers receive large volumes of vague, duplicate, or misrouted complaints without standardized priority scores. Minor maintenance items end up competing with urgent road hazards.
- **Absence of Community Engagement:** Citizens have little motivation to report neighborhood issues, and field teams lack a unified view of problem density across wards.

Motivation: The goal was to demonstrate that a clean web interface, spatial map visualization, and basic automated triage can replace complex ticketing workflows with an open, collaborative system.

3. WHAT WE WANTED TO ACHIEVE

Objective	Definition of Success
1. Low-Friction Intake	Citizens can report an issue in under 60 seconds with location coordinates, photos, and auto-categorization.
2. Automated Incident Triage	Complaints are classified by category and assigned an urgency priority score (0–100) automatically.
3. Spatial GIS Map Visibility	A map interface that displays color-coded incident pins across wards for rapid evaluation by officers.
4. Transparent Progress Loops	Clear status timelines and activity notes showing each stage of ticket resolution.
5. Executive Decision Support	Automated compilation of daily operations summaries and ward-level workload statistics.

4. THE IDEA & APPROACH

Rather than treating complaints as disconnected database records, Jansathi was designed as a connected workflow between two primary roles:

Citizen Finds Issue → Web Submission (Photos & Coordinates) → AI Triage (Category & Priority) → Municipal GIS Queue → Crew Dispatch & Resolution → Citizen Verified & Rewarded

Core Elements of the Approach:

- **Dual-Role Architecture:** Distinct, tailored interfaces for citizens and municipal officers.
- **High-Level AI Assistance:** Automatically extracting issue categories, severity estimates, and text summaries without manual sorting.
- **Geographic Mapping:** Using interactive map layers to show spatial clusters and problem hotspots.
- **Community Reputation:** Rewarding verified reporting with activity milestones and trust points.

5. TECHNOLOGY STACK & IMPLEMENTATION OVERVIEW

Jansathi was engineered using modern web technologies to ensure high performance, responsive mobile accessibility, and clean component isolation. Below is an overview of the technology stack and where each component is utilized:

Technology	Role in Stack	Where & How It Is Used in Jansathi
React 19 & JSX	Frontend UI Framework	Powers the single-page application (SPA), dynamic role routing (Citizen vs. Officer), interactive feed rendering, and real-time form preview state management.
Vite 8	Build Tool & Dev Server	Provides ultra-fast Hot Module Replacement (HMR), asset bundling, and production build optimization for rapid client delivery.
Tailwind CSS v4	Styling & Design System	Used across all dashboards, forms, and navigation for utility-first styling, glassmorphic card overlays, and dynamic light/dark theme transitions.
Leaflet GIS Engine	Geospatial Mapping	Renders interactive vector maps on the Citizen and Officer Dashboards with custom color-coded hazard pins, dynamic bounding boxes, and hover tooltips.
Google Gemini AI API	Multimodal AI Analysis	Analyzes unstructured complaint descriptions, determines urgency priority scores (0–100), extracts category tags, and compiles natural language city briefings.
Firebase Firestore / Auth	Cloud Identity & Data Sync	Provides authenticated session management for citizens and municipal officers, alongside resilient real-time cloud and local mock storage fallbacks.
Lucide React Icons	Iconography	Provides clean, lightweight SVG icons for navigation, status indicators, department badges, and UI action triggers.
React Router DOM v7	Client-Side Routing	Manages navigation between Citizen Feed, Report Issue, Officer Command Center, Profile Guild, and AI Agent Hub.

6. WHAT I BUILT: CORE USER-FACING FEATURES

Below is a summary of the core user-facing features and operational modules developed in Jansathi:

1. Incident Reporting Form with Live Preview

A single-page reporting interface that lets citizens provide descriptions, upload photos, and select locations via an interactive map pin or address search. It displays a live preview card with estimated severity and reward points before submission.

2. Municipal Command Center & GIS Map

A centralized dashboard for municipal staff that displays department statistics, active backlog counts, and an interactive GIS map. Pins on the map indicate incident locations and are color-coded by severity (Critical, High, Medium, Low), allowing officers to inspect details on hover or click.

3. Central Incident Triage Queue

A structured table of all registered civic reports with filters for status, department, and priority score. Officers can review citizen photos, assign responsible divisions, log work notes, and update resolution states.

4. Autonomous AI Intelligence Hub

Background operational modules designed to scan for duplicate complaints, evaluate localized infrastructure risks, and compile automated city briefing summaries.

5. Executive PDF Report Generator

An export engine capable of generating standardized daily operations reports. For example, in sample runs (Ref: JS-OP-67260), the system aggregates total complaints (e.g., 5 total, 4 resolved, 1 pending), calculates department workload shares (Roads 40%, Unassigned 40%, Sanitation 20%), and outputs prioritized strategic recommendations.

6. Citizen Profile, Verification & 24/7 AI Assistant

Features citizen reputation metrics, an identity verification workflow for authentic reporting, and an integrated conversational AI helper to guide users through platform features and civic questions.

7. DID IT ACTUALLY SOLVE THE PROBLEM?

Objective	What Was Built	Observed Result	Status
Low-Friction Intake	Unified form with live preview and map pin locator.	Reports filed in under 60 seconds with clear categorization.	Fully Achieved
Automated Triage	Automated category matching and priority scoring (0–100).	Complaints organized by urgency without manual bottlenecks.	Fully Achieved
Spatial Map Visibility	Interactive map with severity pins and hover inspection.	Officers can evaluate geographic clusters instantly.	Fully Achieved
Transparent Progress Loops	Timeline logs, status cards, and reward scores.	Complete visibility over ticket status from open to resolved.	Fully Achieved
Legacy Backend Sync	Simulated database and client-side service modules.	Functional prototype, but not connected to live government databases.	Partially Achieved

8. OUTCOMES & RESULTS

While Jansathi was built as an independent prototype and has not yet undergone live city-wide deployment, internal testing and user walkthroughs demonstrated positive qualitative results:

- **Intake Usability:** Test users completed structured issue reports in 45–60 seconds without confusion over municipal department routing.
- **Operational Clarity:** Dispatchers could quickly identify urgent reports from the map view and access photos and location data in one click.
- **Reporting Speed:** Administrative operations summaries that previously required manual compilation could be generated instantly in clean printable formats.

9. CHALLENGES I FACED

1. Map Hover Interactions and State Synchronization

Problem: Initially, map pins only opened report details on explicit clicks. Users wanted rapid inspection by hovering over pins.

Difficulty: Third-party map DOM elements run outside the primary React event cycle, leading to state synchronization issues.

Resolution: Added direct mouseover event handlers on marker initialization to update active selections and render lightweight HTML tooltips cleanly.

2. Visual Balance Across Light and Dark Modes

Problem: Certain container elements retained dark background styles when switching to light mode.

Difficulty: Utility class defaults conflicted with dynamic theme switching.

Resolution: Refactored the dashboard components with dedicated responsive theme classes and dynamic tile layer reloading.

3. Repository Secret Sanitization

Problem: A Git push was blocked because an early commit in repository history contained an API key.

Difficulty: The key was in the initial root commit, meaning a standard commit could not remove it.

Resolution: Used Git history filtering to sanitize the file across all commits and replaced it with environment variable placeholders.

10. WHAT DIDN'T WORK

- **Multi-Step Wizards:** An initial prototype used a four-step wizard for filing complaints. User testing showed drop-offs. Switching to a single-screen layout with a live preview card was much faster and simpler.
- **Pure GPS Reliance:** Automatic browser GPS occasionally drifted indoors. Adding an interactive draggable pin with address lookup proved necessary for reliable location accuracy.

11. WHAT I LEARNED

Technical Learnings:

- Managing third-party mapping lifecycles and event teardowns within modern web frameworks.
- Implementing clean theme systems using utility-first styling.
- Git history sanitization and secure credential workflows.

Engineering & Problem-Solving Learnings:

- Designing for simplicity: prioritising single-page workflows over complex multi-step forms.
- Building graceful fallbacks when external services are unavailable.
- Structuring unstructured user text into consistent, actionable data fields.

12. WHAT I WOULD IMPROVE

- **Offline Support:** Enable citizens to draft complaints and capture photos offline, auto-uploading when connectivity is restored.
- **Duplicate Photo Detection:** Compare image features on upload to group multiple reports of the same physical issue automatically.
- **Direct Messaging Alerts:** Send resolution status updates directly through SMS or messaging channels.

13. UPCOMING FEATURES & FUTURE ROADMAP

To expand Jansathi into a complete, enterprise-grade urban management ecosystem, the following upcoming capabilities are planned for upcoming development cycles:

- **1. Multilingual Voice-to-Report Intake:** Integrating AI speech-to-text models supporting regional Indian languages (e.g., Kannada, Hindi, Tamil, Telugu) so non-literate citizens can report issues by simply speaking.

- **2. WhatsApp & Telegram Civic Chatbot:** Allowing citizens to submit photos and live location pins directly over popular messaging apps without needing to install or open a browser portal.
- **3. Municipal ERP & CRM Direct Connectors:** Two-way automated webhook synchronizations with municipal ERP systems (such as SAP Public Sector, e-Nagarpalika, and Smart City Command Centers).
- **4. IoT Sensor Telemetry Integration:** Ingesting automated data feeds from smart city sensors (such as ultrasonic trash can fill-level sensors, water pipeline pressure monitors, and smart streetlight grid telemetry) alongside citizen reports.
- **5. Automated Route Optimizer for Work Crews:** An AI dispatch module that clusters open neighborhood tickets and generates turn-by-turn optimal daily driving routes for asphalt trucks, waste compactors, and utility repair vans.
- **6. Citizen Reward & Civic Credit Marketplace:** A redeemable reward catalog allowing citizens to convert verified community XP points into public bus/metro transit credits, property tax rebates, or discounts at participating local businesses.
- **7. Automated Drone Hazard Verification:** Triggering automated commercial drone path inspections for hazardous or inaccessible areas (e.g., high-voltage transmission lines, deep canal blockages, or collapsed bridges).

14. FINAL REFLECTION

Building **Jansathi** began with a simple premise: civic issues should not disappear into an untracked backlog, and municipal workers should not have to sift through disorganized complaints.

Developing this project provided valuable experience in designing clean user experiences, handling geospatial data, and applying automated triage to make municipal reporting practical, clear, and transparent.